Anurag Vishwanatham

ma23btech11004@iith.ac.in | anurag@vishwanatham.in | +91 7093550707 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.97
FIITJEE Junior College	Class XII (Telangana State Board of Intermediate Education)	2023	97.90%
DDMS.P.Obul Reddy Public School	Class X (Central Board of Secondary Education)	2021	96.80%

ACHIEVEMENTS

- Achieved a **maximum rating** of **1741** in CodeForces (profile)
- Achieved **maximum rating** of **1716** in CodeChef (profile)
- Won **2nd place for Hostel Sarabhai** in **Gridlock**, a competitive programming contest during **Milan 2024**, the annual inter-hostel competition of IIT Hyderabad; participated as part of a **3-member team**

WORK EXPERIENCE

Oracle - Project Intern

- Improved an **AI-powered incident triage agent** used by on-call engineers by investigating gaps reported and implementing fixes across its reasoning and evidence-collection workflows.
- Extended the agent with tools to inspect **Kubernetes pod logs** and retrieve **latest Airflow DAG runs**, enabling automated diagnosis of executor-assignment and transient failures.
- Improved triage reliability by adding **partial-triage labeling** and instructions for identifying incomplete investigations, reducing ambiguity for on-call engineers.
- Implemented **root-cause correlation** across tickets using titles, comments, resolution descriptions, and temporal proximity to identify bursts of incidents with likely common causes.
- Developed a **Cost Analysis Report Generator** that analyzes multi-region and tenancy-wide expenditure, identifies cost trends and anomalies, and generates Markdown/PDF reports with evidence-based cost-saving recommendations.

University of Tokyo - Research Intern

- Implemented the construction of **explicit lossless vertex expanders** described in Hsieh, Lubotzky, Mohanty, Reiner & Zhang, *Explicit Lossless Vertex Expanders* (arXiv:2504.15087).
- Built the construction based on the **product** of a **constant-sized lossless expander** and a base graph arising from **Ramanujan-Cayley cubical complexes**.

PROJECTS

Statistical Analysis of Covid-19 (GitHub)

- **Collected real data** on COVID-19 to analyze the effect of the pandemic across different countries and timelines
- **Visualized data** by creating pie charts, bar graphs, histograms, ogives and box plots
- Verified **Central Limit Theorem** with the data; Estimated gamma distribution parameters using **MLE** and **MOM estimators**
- **Constructed confidence intervals** for variance and mean differences between regions of WHO; Applied **hypothesis testing** to check susceptibility of consumers of tobacco to COVID-19

Reed Muller and Polar Codes: Comparative Decoding (Github)

- Implemented **Maximum Likelihood**, **Successive Cancellation**, and **Majority Logic decoding algorithms** for **Reed-Muller codes** and ML/SC decoding for **Polar codes** over **Binary Symmetric Channels**.
- Evaluated **codeword error probability** across different channel error rates and compared the decoding performance of the two code families.

Implemented Linear Regression using Gradient Descent (GitHub)

- Explored and Implemented **Linear Regression** using **Gradient Descent** from **scratch** in **C++**
- Applied **batch gradient descent** and **mini-batch gradient descent** with optimizations including **Nesterov momentum**, **RMS prop**, **AdaGrad** and **Adam**.

Secure file transfer with AES encryption (GitHub)

- Built a **client-server** model using **python** for secure file transfer using **AES encryption**.
- Utilized **sockets** and **threading** to enable reliable multi-client communication.

SKILLS

Languages: C++, C, Python, SQL, Matlab

Tools: Latex, Github, Numpy, Matplotlib, Qiskit

RELEVANT COURSES

Computer Science: Approximation Algorithms, Algorithms on Large Graphs, Pseudorandomness, Pattern Recognition and Machine Learning, Data Structures and Algorithms, Theory of Computation, Database Management Systems

Mathematics: Spectral Graph theory, Graph theory and Combinatorics, Linear Algebra, Probability theory, Applied Statistics, Groups and Rings, Modules and Fields, Number Theory.

Miscellaneous: Quantum Computing, Quantum Error Correcting Codes, Error Correcting Codes

POSITIONS OF RESPONSIBILITY

Core Member, *Infero - The Algorithmic and Logical Reasoning Club, IIT Hyderabad*

Core Member, *The Otaku Club - Anime and Manga Club, IIT Hyderabad*